

The First Impression: Understanding the impact of Multimodal System Responses on User Behavior in Task-oriented Agents

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ABSTRACT

In task-oriented dialog agents, the initial user experience sets the tone for future interactions and plays a major role in whether the user returns to the system. But with the diversity of human interaction, these systems often struggle with more complex requests that differ significantly from their training data creating an unpleasant and frustrating user experience. In this work, we formulate an alternative and less resource-intensive approach to improving user experience by implicitly conveying possible user action in system responses. We make use of real-world data to test and support our claims, showing that a significant share of the users made use of such information. Additionally, we extend this to product recommendation as a way to anchor the interaction to well-established flows, we then perform data analysis that shows that close to a third of the users tried benefited from such recommendations. We conclude with an overview of the plans for my Ph.D. thesis and how these findings line up with our goal of producing an ideal agent.

CCS CONCEPTS

• **Computing methodologies** → **Natural language generation**; Discourse, dialogue and pragmatics; Intelligent agents; • **Human-centered computing** → Interaction design.

KEYWORDS

task-oriented dialog agents, natural language generation, politeness

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1 INTRODUCTION

Modern task-oriented dialog agents work under the premise that the user is not only able to understand how the agent works but also the system limitations, and that the user will accommodate

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Figure 1: Ideal interaction between a user and an ideal agent. Use placeholder images and names for the products to showcase that this system can be domain agnostic. This example is a representation of the end goal of my Ph.D. thesis.

such limitations. This leads to systems capable of near-perfect interactions, capable of offering the user exactly what they were looking for, but only when operated by near-perfect users.

This assumption creates a big gap in user experience, with a user that emulates the ideal user having an ideal interaction, while a user that strays from that behavior has a very different experience. In a live setting, where users' demographics become highly diverse, it is not feasible to assume that users will be restricted to a pre-defined set of interaction flows. Humans are diverse and with that comes a vast variety of ways to interact with task-oriented dialog agents, creating the need for very large models [2, 4, 7, 18] capable of identifying the user's intention in almost any form that it might

be presented. This however requires either very comprehensive datasets or a constant expansion of the used models as new data is available, making such an approach hard to maintain or scale.

We believe that a less resource-intensive approach could be used in the form of user conditioning. That is, to implicitly encourage the user to make the decision that keeps the interaction on track and is more beneficial for the user experience. This can take several forms such as recommending a product, rather than letting the user aimlessly browse, or encouraging the user to say a command that advances the conversational flow. This can be further complemented with the usage of visual stimuli such as appealing images of the products, as seen in Figure 1, something that has been shown to impact initial impressions [8].

In my Ph.D. thesis, we seek to leverage our unique position, as participants in the Alexa Prize Taskbot Challenge [6] with team TWIZ [5], to explore such an approach and its effectiveness. We use our real-world data to show that users often latch on to the suggested products and commands, and in doing so produce a better user experience. To summarize, I contribute with the following analysis to support my Ph.D. research direction:

- (1) We present the impacts of system responses on the users and the way they interact with the system
- (2) We perform analysis of real data from a currently deployed task-oriented dialog agent that supports our hypothesis
- (3) We discuss research directions that leverage this behavior to improve task-oriented models.

2 RELATED WORK

First impression biases have been carefully studied for the last few decades [1, 12, 14, 19] with work mainly focusing on faces. With the progress of technology and the advent of the internet, textual first impressions are becoming a common occurrence. Lim et al. [8] explored how the presence of a multimedia element on a first impression could mitigate the negative impacts of a biased text-only first impression. Thielsch et al. [17] focuses on evaluating the impact of first impressions on websites, they focus on three aspects: content, perceived usability, and aesthetics. Their work shows that a user's intention to revisit a website is significantly impacted by its initial content, but all three aspects play a part in the user's overall impression. In a more human-focused vein, Mattarozzi et al. [10] conducted an experiment to understand how gender and personality impact first impression. This work concluded that both gender and personality play a part in judging the trustworthiness of a face, especially so when judging trustworthy-looking faces. Despite this, to the best of our knowledge, no work has been done to explicitly reduce these biases.

As for intent detector variability, Stoica et al. [16] while introducing an n intent detector for Romanian they show that one of the biggest challenges was utterance complexity with a 40% drop in performance when synonyms are used. In Ray et al. [13], to address this problem, the authors introduce a paraphrasing model into the pipeline that paraphrases the user utterance onto something that the intent detector might have higher confidence classifying.

Recently some focus has also been put on tone of voice, particularly polite tone. Danescu-Niculescu-Mizil et al. [3] introduced a politeness dataset where sentences were manually annotated for

politeness levels, allowing further research into the subject. This dataset was then used by Niu and Bansal [11] and Madaan et al. [9] where the authors proposed a generative and rewriting approach to introducing politeness into generated answers. Our recent work [15], compared these two approaches under a task-oriented dialog agent scenario, showing that rewriting approaches proved beneficial for system response quality and tone.

3 DATA

For our work, we will be using the data collected during our participation as team TWIZ [5] in the Alexa Prize Taskbot Challenge [6], where users would voluntarily interact with the agent using one of their echo devices. This creates a dialog dataset of real user-agent interactions, where the purpose of each interaction is to assist the user with a recipe or DIY task, from its selection to completion. Furthermore, the users are asked to leave a rating or some feedback, at the end of each interaction, allowing us to relate new features to user satisfaction. This dataset is composed of thousands of interactions, totaling tenths of thousands of user-systems turns. On average, each interaction lasted over 7 turns. For each, we collected the given user rating, the start time, device type, user feedback, duration, and unique user and conversation identifiers. For the turns, we also collected the date, detected intent, and conversation phase.

4 IMPACT OF SYSTEM RESPONSES

In this paper, we study the impact of system responses on the user input and how user-guiding system responses can be used to produce improved user-agent interactions. To do this, we support ourselves in the data collected during the year-long deployment of a task-oriented dialog agent under the Alexa Prize Taskbot Challenge. This data is described in Section 3.

Over the next sections, we will discuss some hypotheses followed by data-backed analysis to help understand the impact of system responses on several key aspects of user input.

4.1 Impact on User Utterance

When dealing with traditional task-oriented dialog agents, users need to be made aware of the possible commands and ways to interact with the bot. One such example is how traditional Alexa skills behave, supporting a very limited amount of possible commands and with little variability, in the way they are invoked (e.g. a skill can support the utterance *"Add X to my shopping list"*, but it might struggle with the utterance *"Could you please include X in the list of things to buy"* even though they carry the same intention).

While, ideally, modern task-oriented dialog agents should support any variation of the same command, when dealing with real users the possibilities are endless, pushing systems to adopt a *learn-as-you-go* approach, where dialog managers and intent detectors are periodically re-trained with new conversations to improve accuracy.

This limitation not only makes the system hard to scale but also exposes the system's unintelligent behavior and worsens the early user experience. Early in TWIZ's development cycle, we observed that users, when unsure what to say, would interact with the bot the same way it interacted with them using similar vocabulary and expressions. From this we drew two main hypotheses:

Measure	Similarity (%)
Sim	45.28
Sim+StopWords	35.47
Sim+StopWords+Stemming	36.10

Table 1: Average similarity between the user utterance and the preceding system response.

- (1) We can **simplify and condition user input** by implicitly or explicitly telling the user what to say
- (2) Inversely, we could provoke **more varied and complex user input** by omitting instructions and commands from system responses.

To analyze the first point, we performed n-gram similarity computations over the user utterance and the previous system response. Specifically, we extracted all system response - user utterance pairs, and then we measured the percentage of shared uni-grams between the user utterance and the system response. This metric works on the premise that, over a large enough pool of users, several ways of saying the same thing will emerge, therefore uninfluenced input would have low scores on average. After excluding any single word user utterance, we analyzed around 96% of interactions (the remaining 4% did not have any turns that match the criteria) accounting for 45% of possible pairs.

Table 1, shows the results of this analysis when measuring similarity while removing stop-words (SW) and stemming (STEM) the utterances. In this table, we see that, on average, more than a third of the user utterance is present in the preceding system response, showing that answer generation models for task-oriented dialog agents could improve user experience if the generation is conditioned to include some possible user commands.

This first point can be further visualized in the system responses from Figure 1. In this scenario, the interaction progresses naturally but the system proposes alternative courses of action to the user, without explicitly saying "Say X to do Y". This is a key feature of what we believe to be an ideal agent, as it exposes the system functionalities in a natural manner. This was something employed in TWIZ's responses to reduce the user's need to explicitly ask the bot what to do, which would lead to a more artificial and undesirable flow.

As for the second point, while we were able to verify it when manually analyzing each conversation, we also noted that this particular scenario was not frequent enough to perform any in-depth analysis.

4.2 Impact on Product Selection

One of the distinctive characteristics of TWIZ was that it offered DIY tasks and recipe suggestions upon opening the taskbot. This was done in an effort to guide the interaction to favorable flows. In this section, we will perform a similar analysis to Section 4.1, to understand how often were the users influenced by the taskbot's initial suggestions.

Here we looked into the three suggestions we make initially (one recipe, one DIY task, and the seasonal suggestions), we measured the percentage of uni-grams that overlap each suggestion and the

user utterance and then consider only the highest. We consider that the user was influenced by the suggestion if more than half of a suggestion uni-grams are present in the user's input. Table 2, highlights an example of a system response - user utterance pair to clarify what constitutes a suggestion and how we measure influence.

Through our analysis, we see that in 29.2% of the cases the user was influenced by our suggestions. Furthermore, we also observed a 4.7% average rating improvement in said cases and a 6.3% increase in the average interaction length. This shows that, in task-oriented dialog agents, not only it is beneficial to provide users with an initial product selection but also that a curated initial selection could lead to improved customer experience and retention. We believe that this could be attributed to the fact that a significant portion of first-time interactions is led by curiosity rather than need, this is, users, engage with the system to explore it and its functionalities before coming back in with a genuine need for help with DIY task or recipe.

Looking back at Figure 1, we see this principle in action with several options being offered to the user in the first turn. This example also mimics a feature implemented in TWIZ, using images to help lead the user towards one of the suggested interaction flows. Using images this way can be especially useful in domains such as food or fashion, where two products with the same name can be very different visually. Due to its nature, we could not evaluate the impact of the presented images in the user's selection.

5 WORK IN PROGRESS

The research presented here is part of the end goal of my Ph.D. thesis, which is to produce a domain-aware empathetic natural task-oriented dialog agent, similar to the one presented in Figure 1. To this end we will be using this research to condition the generation on a set of products or commands, to ensure that the model offers them to the user. In particular, we will be using a multimodal text realization model, rather than a text generation model. This type of model produces a delexicalized template response, rather than a complete response, that can then be filled with attribute values. To include the commands and recommended products we will treat them as attributes that can be added to the response, increasing the model's flexibility.

To make the agent empathetic will be using reinforcement learning techniques paired with a sentiment analysis model. The goal here is for the generation to be guided by the sentiment detected on the user input allowing the agent to acknowledge and adjust to the user's mood. An example of this can be seen in the third system response of Figure 1 where the system takes notice of the user's declared mood and incorporates that into its response.

Furthermore, we are producing a dataset for the task of Politeness introduction. This dataset is an important step as it targets the main limitation of the existing dataset (see 2) which is domain dependency. The data will then be used to produce a rewriting politeness model that rewrites the system responses into polite versions of themselves, to help improve user experience. This model follows a similar architecture to that of Tag & Generate [9] but, based on our previous findings [15], we will redo the Tagger component. In

System Response	Hi, this is an alexa prize taskbot. Happy holidays! You can ask me for recipes like, 'how to bake a cake', or a DIY task such as 'how to paint a wall'. Or better yet, try our holidays suggestions!
User Utterance	What are your holiday suggestions?
Suggestion 1	bake a cake
Suggestion 2	paint a wall
Suggestion 3	holidays suggestions

Table 2: Example of how decided if a given user utterance was influenced by our initial suggestions. We note that the third suggestion is completely contained in the user utterance, meaning that it was influenced by it.

particular, we will tackle the style marker selection algorithm so that it becomes style-specific (the polite style in our case).

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